Solutions

Answer each question and upload your answers using the link in Blackboard. Each question is worth the number of marks indicated on the right.

1. (a) Translate the following argument into symbolic form, using the specified statement variables.

* Let p be "It is hot"
* Let q be "It is cloudy"
* Let r be "It is raining"
* Let s be "It is sunny".

Argument: *It is hot and not sunny. Being cloudy is necessary for it to be raining. It is either raining or sunny, but not both. Therefore, it is cloudy.*

(2 marks)

$p \wedge \sim s$

$\sim q \rightarrow \sim r$

$(r \vee s) \wedge \sim (r \wedge s)$

$\therefore q$

(b) Determine whether the argument is valid or invalid. Show your working. **(4 marks)**

Solution 1 Suppose the argument is invalid, so each of the premises is true and the conclusion is false.

Then $\boxed{q \text{ is false}}$.

Since q is false and premise 2 is true, $\sim r$ must be true so $\boxed{r \text{ is false}}$.

Hence since premise 3 is true, $\boxed{s \text{ is true}}$.

But this contradicts the fact that premise 1 is true.

Thus the argument cannot be invalid, so it is valid.

Solution 2

1. $p \wedge \sim s$
2. $\sim q \rightarrow \sim r$
3. $(r \vee s) \wedge \sim (r \wedge s)$
4. $\sim s$ from 1 by Specialisation
5. $r \vee s$ from 3 by Specialisation
6. r from 4 and 5 by Elimination
7. $\sim(\sim q)$ from 2 and 6 by Modus Tollens
8. q from 7 by double negative law

$\therefore$ The argument is valid.

2. **(a)** Prove the following statement:

$$\forall m, x \in \mathbb{R}, \text{ if } m \in \mathbb{Z} \text{ and } x \notin \mathbb{Z}, \text{ then } \lceil x \rceil + \lceil 2m - x \rceil = 2m + 1.$$

(4 marks)

Suppose $m, x \in \mathbb{R}$, $m \in \mathbb{Z}$ and $x \notin \mathbb{Z}$.

Since $x \notin \mathbb{Z}$, $\exists\, a \in \mathbb{Z}$ such that

$$a < x < a+1.$$

Thus $\lceil x \rceil = a+1$.

Also $-a > -x > -(a+1)$

so $2m-a > 2m-x > 2m-a-1$.

Since $m, a \in \mathbb{Z}$ $2m-a$ and $2m-a-1 \in \mathbb{Z}$

so $\lceil 2m-x \rceil = 2m-a$

Hence $\lceil x \rceil + \lceil 2m-x \rceil = a+1 + 2m-a = 2m+1$. $\square$

(b) Disprove the following statement:

$$\forall a, b \in \mathbb{Z}, \quad \gcd(a,b)\text{lcm}(a,b) = ab.$$

(2 marks)

Consider $a = -2$ $b = 4$.

Then $a, b \in \mathbb{Z}$

$\gcd(a,b) = 2$

$\text{lcm}(a,b) = 4$

But $\gcd(a,b)\,\text{lcm}(a,b) = 8$

and $ab = -8$.

Hence the statement is false.

3. Define the sequence $\{a_n\}_{n\geq 0}$ by

$$a_0 = 1, \quad a_1 = 3, \quad \text{and} \quad a_k = 2a_{k-1} + 6a_{k-2} \text{ for each integer } k \geq 2.$$

Use (strong) mathematical induction to prove that $a_n \geq 3^n$ for each integer $n \geq 0$.

(5 marks)

Let $P(n)$ be the predicate $a_n \geq 3^n$.

Since $a_0 = 1$ and $3^0 = 1$, $a_0 \geq 3^0$ so $P(0)$ is true.

Since $a_1 = 3$ and $3^1 = 3$, $a_1 \geq 3^1$ so $P(1)$ is true.

Suppose that for some $k \in \mathbb{Z}$, $k \geq 2$, $P(0), P(1), \ldots P(k-1)$ are all true.

$$\begin{aligned}
a_k &= 2a_{k-1} + 6a_{k-2} && \text{as defined in the question} \\
&\geq 2 \cdot 3^{k-1} + 6 \cdot 3^{k-2} && \text{by the inductive hypothesis} \\
&= 2 \cdot 3^{k-1} + 2 \cdot 3 \cdot 3^{k-2} \\
&= 2 \cdot 3^{k-1} + 2 \cdot 3^{k-1} \\
&= 3^{k-1} \cdot 4 \\
&\geq 3^{k-1} \cdot 3 \\
&= 3^k
\end{aligned}$$

Thus $P(k)$ is true.

Hence by the PSMI, $a_n \geq 3^n$ for each integer $n \geq 0$.

4. (a) For each of the following statements, indicate whether the statement is true or false by circling the appropriate response.

(2 marks)

$\mathbb{Z} \in \mathcal{P}(\mathbb{Q})$ True False

$\mathbb{Z} \subseteq \mathcal{P}(\mathbb{Q})$ True False

$\emptyset \in \mathcal{P}(\mathbb{Q})$ True False

$\emptyset \subseteq \mathcal{P}(\mathbb{Q})$ True False

(b) Prove the following statement.
For all sets A, B that are subsets of a universal set U, $(A - B) \cup (A \cap B) = A$.

(4 marks)

$(A - B) \cup (A \cap B)$

$= (A \cap B^c) \cup (A \cap B)$ by the Set difference law

$= A \cap (B^c \cup B)$ by the Distributive law

$= A \cap U$ by the complement law

$= A$ by the identity law

$\therefore (A - B) \cup (A \cap B) = A$.

5. Consider the function $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \times \mathbb{Z}^+ \to \mathbb{Z}^+$ where $\forall (a, b, c) \in \mathbb{Z}^+ \times \mathbb{Z}^+ \times \mathbb{Z}^+$, $f((a, b, c)) = abc$.

(a) State the image of $(4, 6, 1)$, that is, $f((4, 6, 1))$. **(1 mark)**

$f((4,6,1)) = 24$

(b) State the pre-image of 3, that is $f^{-1}(\{3\})$. **(1 mark)**

$f^{-1}(\{3\}) = \{(3,1,1), (1,3,1), (1,1,3)\}$

(c) Is f one-one? Explain your answer. **(2 marks)**

No.

$f((3,1,1)) = 3$ and $f((1,3,1)) = 3$

but $(3,1,1) \neq (1,3,1)$

(d) Is f onto? Explain your answer. **(2 marks)**

Yes.

Consider $x \in \mathbb{Z}^+$.

Then $(x,1,1) \in \mathbb{Z}^+ \times \mathbb{Z}^+ \times \mathbb{Z}^+$

and $f((x,1,1)) = x \cdot 1 \cdot 1 = x$.

6. Define ~~Create~~ a function that demonstrates that $|\mathbb{Q}| \leq |\mathbb{R}^+|$. You must explain why ~~prove that~~ it is a function and explain why it shows that $|\mathbb{Q}| \leq |\mathbb{R}^+|$.

(5 marks)

Let $f: \mathbb{Q} \to \mathbb{R}^+$ where $f(r) = \begin{cases} r+1 & \text{if } r \geq 0 \\ -\sqrt{2}\,r & \text{if } r < 0. \end{cases}$

f is a function since each rational number r is mapped to a unique positive real number, $r+1 \in \mathbb{R}^+$ for $r \in \mathbb{Q}$, $r \geq 0$ and $-\sqrt{2}r \in \mathbb{R}^+$ if $r \in \mathbb{Q}, r < 0$.

f is injective.

Suppose $\exists r_1, r_2 \in \mathbb{Q}$ $r_1 \neq r_2$ s.t. $f(r_1) = f(r_2)$.

If $r_1, r_2 \geq 0$ $f(r_1) = f(r_2) \to r_1 + 1 = r_2 + 1$ so $r_1 = r_2$.

If $r_1 \geq 0, r_2 < 0$ then $f(r_1)$ is rational and $f(r_2)$ is irrational so they cannot be equal.

Similarly if $r_1 < 0$, $r_2 \geq 0$.

If $r_1, r_2 < 0$ $f(r_1) = f(r_2) \to -\sqrt{2}r_1 = -\sqrt{2}r_2$ so $r_1 = r_2$.

So we have a contradiction and

Thus f is injective so $|\mathbb{Q}| \leq |\mathbb{R}^+|$.

7. ~~Let $S = \{1, 2, 3\}$~~. Define a relation ρ on $\cancel{S \times S}$ $\mathbb{Z} \times \mathbb{Z}$ by

$$\forall (a, b), (c, d) \in \cancel{S \times S}\ \mathbb{Z}\times\mathbb{Z},\ (a, b)\rho(c, d) \text{ if and only if } a \le c \text{ and } b \le d.$$

(a) Prove that ρ is a partial order. ~~but not a total order~~. (5 marks)

$\forall (a,b) \in \mathbb{Z}\times\mathbb{Z}$ $a \le a$ and $b \le b$
so $(a,b)\rho(a,b)$.
Thus ρ is reflexive.

Suppose $(a,b) \in \mathbb{Z}\times\mathbb{Z}$, $(c,d) \in \mathbb{Z}\times\mathbb{Z}$, $(a,b)\rho(c,d)$ and $(c,d)\rho(a,b)$.
Then $a \le c$ ①, $b \le d$ ②, $c \le a$ ③, $d \le b$ ④.
From ① and ③ $a = c$, from ② and ④ $b = d$.
$\therefore (a,b) = (c,d)$
Hence ρ is anti-symmetric.

Suppose $(a,b), (c,d), (e,f) \in \mathbb{Z}\times\mathbb{Z}$ and $(a,b)\rho(c,d)$ and $(c,d)\rho(e,f)$.
Then $a \le c$ ①, $b \le d$ ②, $c \le e$ ③, $d \le f$ ④.
From ① and ③ $a \le e$, from ② and ④ $b \le f$.
Hence $(a,b)\rho(e,f)$ so ρ is transitive.
Thus ρ is a partial order.

(b) Prove that ρ is not a total order. (1 mark)

$(1,2) \in \mathbb{Z}\times\mathbb{Z}$ and $(2,1) \in \mathbb{Z}\times\mathbb{Z}$.
But $(1,2) \not\rho\, (2,1)$ since $2 \not\le 1$
and $(2,1) \not\rho\, (1,2)$ since $2 \not\le 1$
so not every pair of elements are comparable
and so ρ is not a total order.

8. Let $G = \{[1], [5], [7], [11]\}$, where $[a] = \{x \in \mathbb{Z} : x \equiv a \pmod{12}\}$.

(a) Draw the Cayley table for $(G, \cdot)$ where $\cdot$ is the operation of multiplication modulo 12.

(2 marks)

$\cdot$	[1]	[5]	[7]	[11]
[1]	[1]	[5]	[7]	[11]
[5]	[5]	[1]	[11]	[7]
[7]	[7]	[11]	[1]	[5]
[11]	[11]	[7]	[5]	[1]

ok if they drop the brackets [ ]

(b) Use your Cayley table to prove that $(G, \cdot)$ is a group.
Note that you may assume that the operation $\cdot$ is associative.

(3 marks)

The body of the Cayley table contains only the elements $[1], [5], [7], [11]$ so the set $(G, \cdot)$ is closed.

$\forall [a] \in G \quad [1] \cdot [a] = [a] \cdot [1] = [a]$
so $[1]$ is the identity of $(G, \cdot)$.

The identity occurs once in each row and once in each column and this shows that inverses exist:

$[1]^{-1} = [1] \quad [5]^{-1} = [5] \quad [7]^{-1} = [7]$ and $[11]^{-1} = [11]$

$\therefore (G, \cdot)$ is a group.

(c) From class we know that $(\mathbb{Z}_4, +)$ and $(\mathbb{Z}_2 \times \mathbb{Z}_2, +)$ are two non-isomorphic groups that each have four elements. Which one of these groups is isomorphic to $(G, \cdot)$? Explain your answer briefly.

(1 mark)

$(\mathbb{Z}_2 \times \mathbb{Z}_2, +)$ since in both groups each element is self-inverse.

9. Solve the equation $11x + 10 = 5$ in the field $(\mathbb{Z}_{19}, +, \cdot)$. Hence determine the smallest positive integer y such that $11y + 10 \equiv 5 \pmod{19}$.

(3 marks)

In $\mathbb{Z}_{19}$

$$11x + 10 = 5$$

$$11x = -5$$

$$7 \cdot 11x = 7 \cdot (-5)$$

$$x = -35$$

$11^{-1} = 7$ since $77 = 4 \cdot 19 + 1$

Since $-35 \equiv 3 \pmod{19}$

$\boxed{y = 3}$

\# primes 7 \# even 9
\# composite 10 \# odd 8

10. Let $S = \{\underline{2}, \underline{3}, 4, \underline{5}, 6, \underline{7}, 8, 9, 10, \underline{11}, 12, \underline{13}, 14, 15, 16, \underline{17}, 18\}$.

(a) In how many ways can a set A of 5 integers be chosen from the set S such that A contains exactly two prime numbers?

(2 marks)

Choose 2 primes and 3 composites.

$$\binom{7}{2}\binom{10}{3}$$

(b) In how many ways can a set B of 5 integers be chosen from the set S such that B contains at most two prime numbers?

(2 marks)

0 prime numbers + 1 prime number + 2 prime numbers

$$\binom{10}{5} + \binom{7}{1}\binom{10}{4} + \binom{7}{2}\binom{10}{3}$$

(c) In how many ways can a set C of 5 integers be chosen from the set S such that C contains exactly two prime numbers or exactly two even numbers? (Note that "or" is inclusive or.)

(4 marks)

Let A be the subsets with exactly 2 prime numbers.
Let B be the subsets with exactly 2 even numbers.
We need $|A \cup B|$.

$$|A \cup B| = |A| + |B| - |A \cap B|$$

To count subsets that have both exactly 2 prime numbers and exactly two even number we use cases depending on whether or not 2 is chosen.

	even	odd
prime	1	6
comp	8	2

Case 1 Choose 2, 1 other prime, 1 other even, 2 odd composites

$$1 \cdot \binom{6}{1} \cdot \binom{8}{1} \cdot \binom{2}{2} = 6 \cdot 8 = 48$$

Case 2 Choose 2 odd primes, 2 even composites, 1 odd composite

$$\binom{6}{2}\binom{8}{2}\binom{2}{1} = 15 \cdot 28 \cdot 2$$

$$\therefore |A \cup B| = \binom{7}{2}\binom{10}{3} + \binom{9}{2}\binom{8}{3} - (48 + 15 \cdot 28 \cdot 2)$$

11. Let S be the set of positive integers from 1 to 100, $S = \{1, 2, \ldots, 100\}$. Determine, with proof, the largest number of integers that can be chosen from S so that no three of the chosen integers are equivalent modulo 9.

(5 marks)

Partition S into collections of integers that are congruent modulo 9.

$\{1, 10, \ldots, 100\}$, $\{2, 11, \ldots 92\}$, $\{3, 12, \ldots, 93\}$, $\{4, 13, \ldots, 94\}$, $\{5, 14, .. 95\}$, $\{6, 15, .., 96\}$, $\{7, 16, \ldots, 97\}$ $\{8, 17, \ldots 98\}$, $\{9, 18, \ldots, 99\}$.

There are 9 parts in this partition.

If we choose 19 integers from S, then by the pigeonhole principle, 3 of those integers must be from the same part of the partition, an thus would be equivalent modulo 9.

However, we can choose the following 18 integers

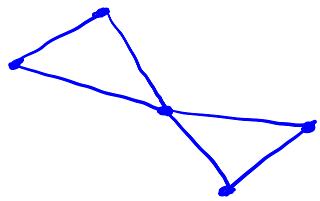
$1, 2, 3, \ldots, 18$ and there are no 3 that are equivalent modulo 9 since there are exactly 2 that are congruent to $x \pmod 9$ for each $x = 0, 1, 2, \ldots 8$.

Thus 18 is the largest number of elements that can be selected from S.

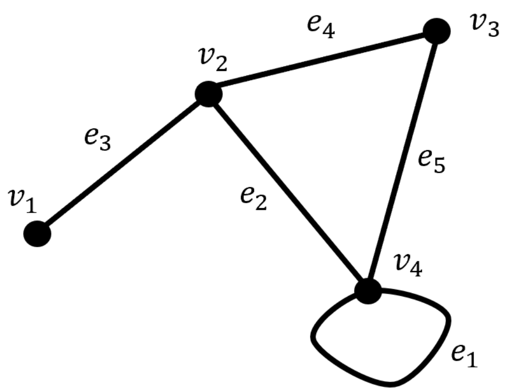
12. **(a)** Draw a graph that has the following adjacency matrix.

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

(2 marks)



(b) State the incidence matrix for the following graph.



$$\begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{ccccc} e_1 & e_2 & e_3 & e_4 & e_5 \end{array} \\ \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 2 & 1 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

(2 marks)

13. Let T be a tree with exactly one vertex of degree 10, exactly two vertices of degree 7, exactly two vertices of degree 3, and in which all the remaining vertices are of degree 1. Determine the number of vertices in T.

Use one or more theorems from the course to

(4 marks)

Let the number of vertices of degree 1 be x.

Thus T has $5+x$ vertices.

Since T is a tree, it must have $4+x$ edges.

$\therefore \quad \sum \deg = 2^{\#}\text{edges}$

$$10+7+7+3+3+x = 2(4+x)$$

$$30+x = 8+2x$$

$$22 = x$$

Thus T has 27 vertices.

END OF EXAMINATION